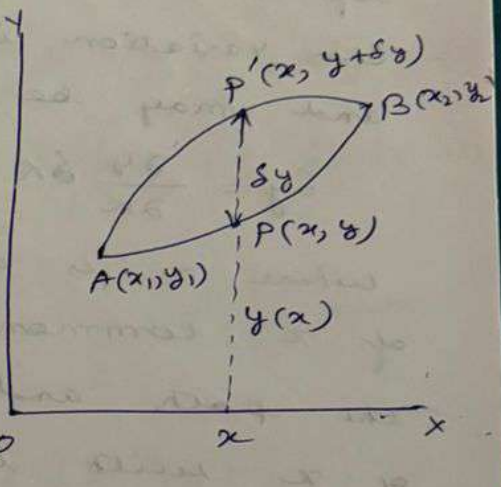
Calculus of

Variational principle and Euler Lagrange's equation:

Let us consider a function $f(y, y', x)$ defined on a curve given by $y = y(x)$ — (1) between two points $A(x_1, y_1)$ and $B(x_2, y_2)$.



Here $y' = \frac{dy}{dx}$. We want to find a particular curve $y(x)$ for which the line integral I of the function f between two points

$$I = \int_{x_1}^{x_2} f(y, y', x)\, dx \quad \text{— (2)}$$

has a stationary value.

Let the curve APB be stationary the curve for which I is stationary. Let us now consider a neighbouring curve AP'B with the same end points A and B. The point P(x, y) of the curve APB corresponds to the point P'(x, y+δy) keeping the x

co-ordinate of the points fixed. This defines the δ-variation of the curve. This variation is arbitary but small and may be expressed as

$$\delta y = \frac{\partial y}{\partial \alpha} \delta\alpha = \eta(x)\,\delta\alpha \quad \text{— (3)}$$

where α is a parameter, independent of x, common to all points of the path and $\eta(x)$ is a function of x with the condition that

$$\delta y_1 = \delta y_2 = \eta(x_1) = \eta(x_2) = 0 \quad \text{— (4)}$$

Now, from (3), the corresponding variation in y' is

$$\delta y' = \eta'(x)\,\delta\alpha \quad \text{— (5)}$$

Now, the integral on the varied path is

$$I' = \int_{x_1}^{x_2} f(y + \delta y, y' + \delta y', x)\, dx$$

$$= \int_{x_1}^{x_2} f(y + \eta\,\delta\alpha, y' + \eta'\delta\alpha, x)\, dx \quad \text{— (6)}$$

Since the variation is small, the integral I' may be obtained by

considering only first order terms in the Taylor expression of the fun^{n} of $\bar{v}$,

$$I' = \int_{x_1}^{x_2} \left[f(y, y', x) + \frac{\partial f}{\partial y} \eta \delta\alpha + \frac{\partial f}{\partial y'} \eta' \delta\alpha \right] dx \quad -(7)$$

Hence $\delta I = I' - I = \delta\alpha \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial y} \eta + \frac{\partial f}{\partial y'} \eta' \right] dx \quad -(8)$

But $\int_{x_1}^{x_2} \frac{\partial f}{\partial y'} \eta' \, dx = \frac{\partial f}{\partial y'} \eta \Big|_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) \eta \, dx$

$$= - \int_{x_1}^{x_2} \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) \eta \, dx \quad \left| \because \eta(x_1) = \eta(x_2) = 0 \right.$$

$$\therefore \delta I = \delta\alpha \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial y} \eta - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) \right] \eta \, dx \quad -(9)$$

The condition that δI is stationary means that $\delta I = 0$ i.e.,

$$\int_{x_1}^{x_2} \left[\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) \right] \eta \, dx = 0 \quad -(10)$$

$\because \eta$ is arbitary, the integrand of (10) must be zero i.e.,

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0 \quad -(11)$$

Eq^{n} (11) is known as the Euler-

Lagrange's equation

If f is a fun^{n} of many independent variables y_k and their derivatives y_k', then we can write

$$\delta I = \delta \int_{x_1}^{x_2} f(y_1, y_2 \cdots y_k, \cdots y_n, y_1' \cdots y_k' \cdots y_n', x)\, dx = 0 \quad \text{---(12)}$$

This leads to

$$\frac{\partial f}{\partial y_k} - \frac{d}{dx}\left(\frac{\partial f}{\partial y_k'}\right) = 0 \quad \text{---(13)}$$

Equ^{n} (13) is the generalized Euler - Lagrange's equ^{n}

Hamilton's principle and Lagrange's equ^{n}

If we identify the Lagrangian to be such that

$$L = L(q_1, q_2 \cdots q_n, \dot{q}_1, \dot{q}_2 \cdots \dot{q}_n, t) \quad \text{---(14)}$$

as compared to the fun^{n}

$$f = f(y_1, y_2 \cdots y_n, y_1', y_2' \cdots y_k', x)$$

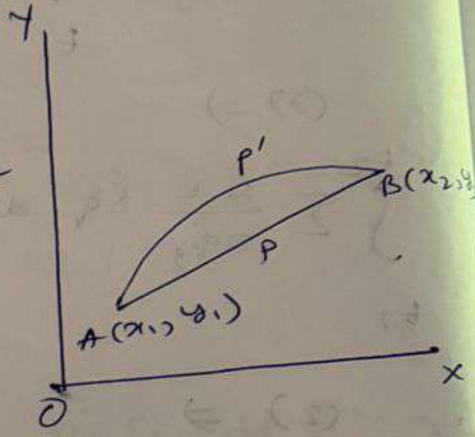
with the transformation

$$x \to t, \quad y_k \to q_n, \quad y_k' \to \dot{q}_n,$$

then the integral $I \to S$, where S is the action integral; then

$$S = \int_{t_1}^{t_2} L\, dt \quad \text{---(15)}$$

Show that the shortest distance between two points in a plane is a straight line.



Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be two points in $x-y$ plane. An element of length ds of any curve say APB passing through A and B is given by

$$ds^2 = dx^2 + dy^2$$

$$\Rightarrow ds = \sqrt{dx^2 + dy^2}$$

$$= \sqrt{\left(1 + \left(\frac{dy}{dx}\right)^2\right)} dx = \sqrt{1 + y'^2}\, dx \quad \text{—(1)}$$

Total length of the curve from A to B is given by

$$S = \int_A^B \sqrt{1 + y'^2}\, dx = \int_A^B f\, dx \quad \text{—(2)}$$

where $f = \sqrt{1 + y'^2}$.

The length of the curve will be minimum when

$\delta s = 0$. This means that f shou

satisfy the Euler - Lagrange eqn

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0 \quad \text{——(3)}$$

Here $\frac{\partial f}{\partial y} = 0$

and $\frac{\partial}{\partial y'} f = \frac{\partial}{\partial y'} (1 + y'^2)^{1/2} = \frac{1}{2}(1+y'^2)^{-1/2} \cdot 2y'$

$$= \frac{y'}{\sqrt{1+y'^2}}$$

$\therefore$ (3) $\Rightarrow$ $\frac{d}{dx}\left[\frac{y'}{\sqrt{1+y'^2}}\right] = 0$, which gives

$\Rightarrow \frac{y'}{\sqrt{1+y'^2}} = \text{const.} = C$

$\Rightarrow y'^2 = C^2(1+y'^2) \Rightarrow y'^2(1-C^2) = C^2$

$\Rightarrow y' = \frac{C}{\sqrt{1-C^2}} = a$ (a constant)

or $\frac{dy}{dx} = a$ —(4), which on integration gives $y = ax + b$, where b is a constant of integration Eqn (4) represents a straight line. Therefore the shortest distance between any two points in a plane is a straight line.